

3.5 Exponential Functions

Exponential functions are used to model situations in which a quantity grows or decreases by a constant percentage over time. They arise naturally in engineering, finance, population growth, radioactive decay, and many other real-world applications.

Definition

An **exponential function** is a function in which the independent variable appears as a power. A general exponential function can be written as

$$y = Ab^{kx}$$

where A is a constant coefficient, b is the base, k controls the rate of growth or decay and x is the independent variable.

In engineering and science, the most important exponential function uses the mathematical constant

$$e \approx 2.71828$$

Definition

A common form of the exponential function is

$$y = Ae^{Bx} + C$$

where A , B and C are constants. When $x = 0$,

$$y = Ae^0 + C = A + C,$$

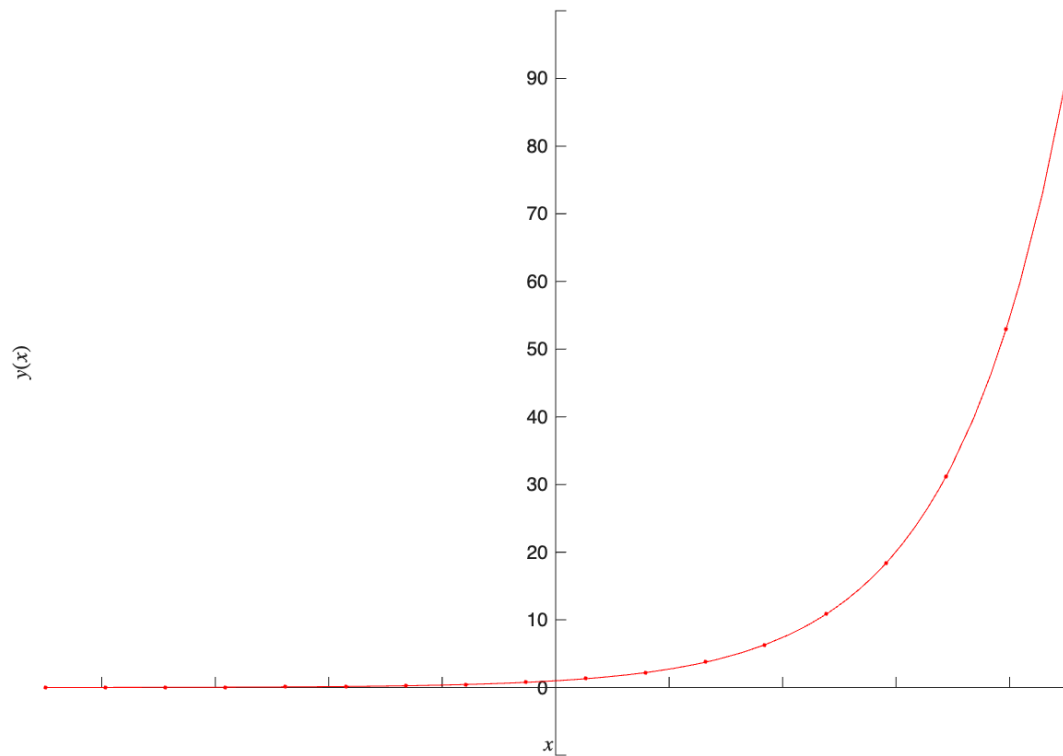
so the y -intercept is equal to $A + C$.

3.5.1 The Graph of an Exponential Function

The simplest exponential function is

$$y = e^x$$

Its graph has the following shape.



Definition

The exponential function

$$y = e^x$$

has the following properties:

- The graph passes through $(0, 1)$.
- The domain is all real numbers.
- The range is $y > 0$.
- The graph never crosses the x -axis.
- The x -axis is a horizontal asymptote.
- The function increases as x increases.

3.5.2 Transformations of Exponential Functions

The constants in

$$y = A e^{Bx} + C$$

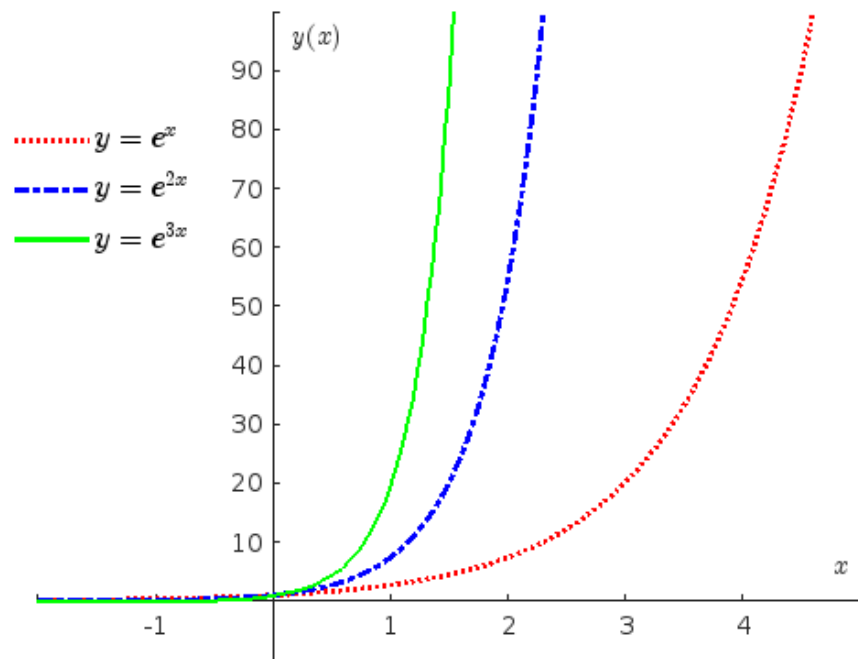
affect the shape and position of the graph.

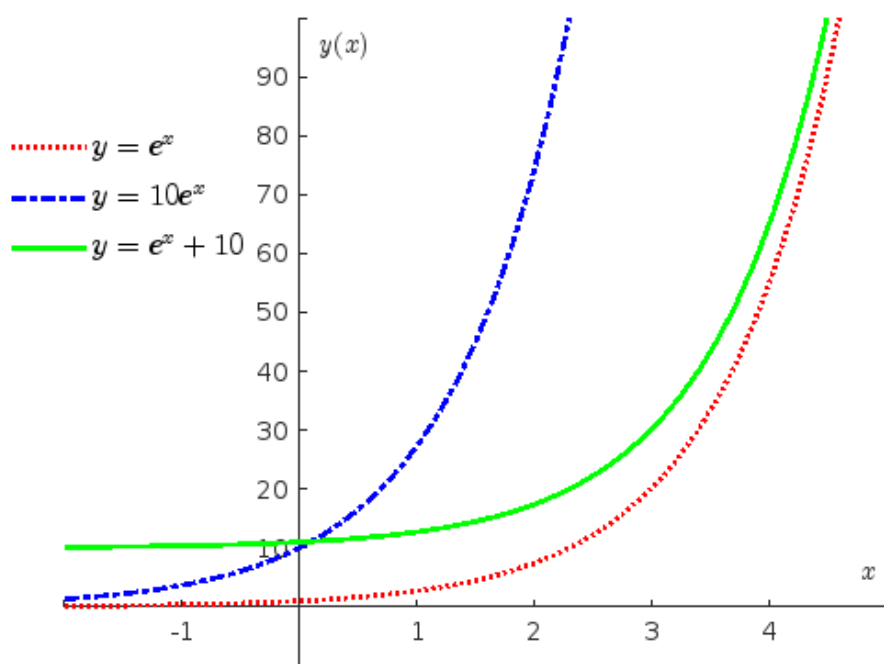
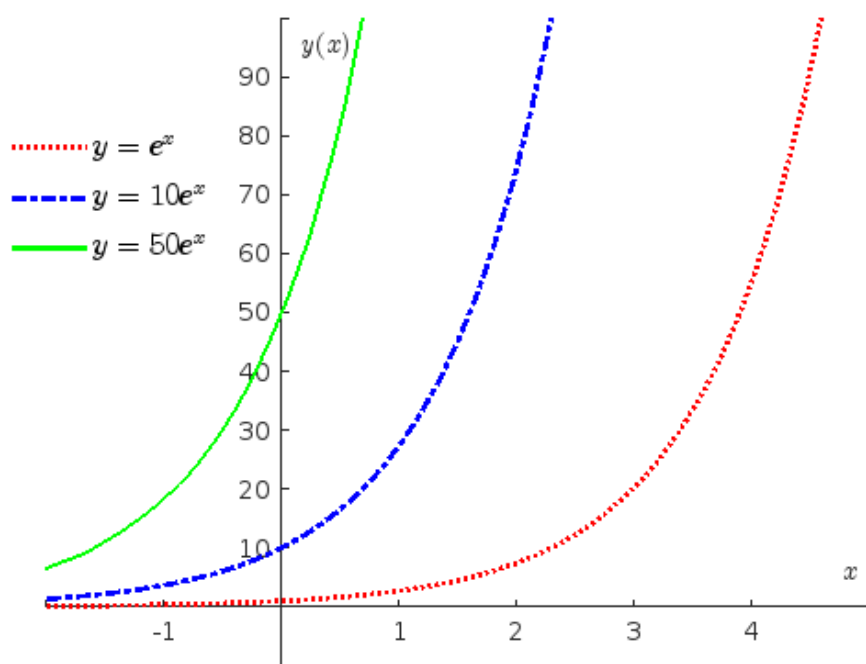
Definition

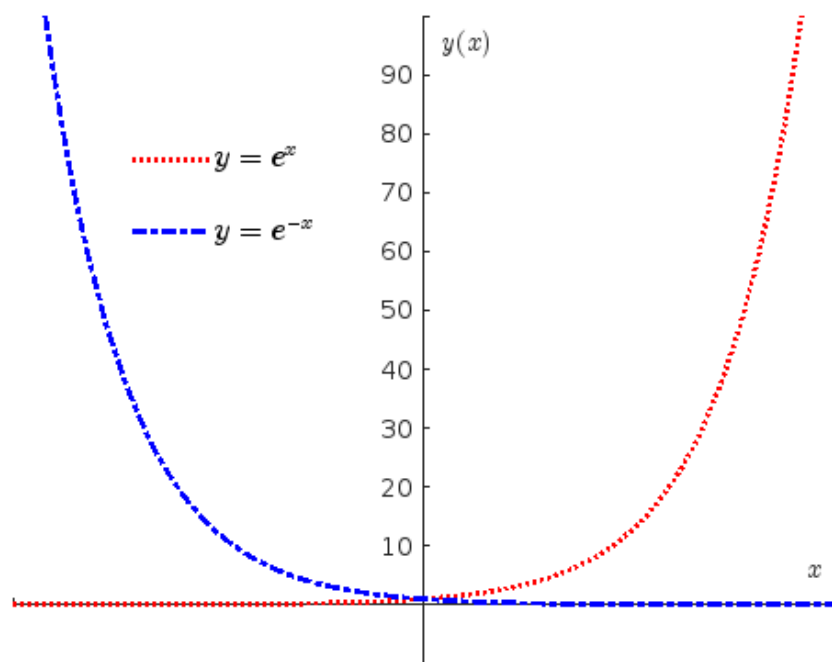
For the function

$$y = A e^{Bx} + C,$$

- Increasing $|A|$ stretches the graph vertically.
- The value of $A + C$ determines the y -intercept.
- Increasing $|B|$ makes the graph steeper.
- If $B > 0$, the function exhibits exponential growth.
- If $B < 0$, the function exhibits exponential decay.
- The constant C shifts the graph vertically.







3.5.3 Evaluating Exponential Functions

Scientific calculators contain a dedicated key for evaluating powers of e .

Worked Example

Evaluate

$$e^4$$

Using a calculator,

$$e^4 \approx 54.60$$

Worked Example

Evaluate

$$4e^{7.2}$$

Using a calculator,

$$4e^{7.2} \approx 4(1339.43)$$

$$\approx 5357.72$$

3.5.4 Solving Simple Exponential Equations

If an equation contains an exponential term, logarithms can be used to determine the unknown power.

Worked Example

Solve

$$e^x = 12$$

Taking natural logarithms of both sides:

$$e^x = 12$$

$$\ln(e^x) = \ln(12)$$

$$x = \ln(12)$$

$$= 2.485 \text{ to 3 decimal places}$$

Worked Example

Solve

$$4e^{3x} + 5 = 12$$

Subtracting 5 from both sides gives

$$4e^{3x} = 7$$

Dividing by 4:

$$e^{3x} = \frac{7}{4}$$

Taking natural logarithms:

$$\ln(e^{3x}) = \ln\left(\frac{7}{4}\right)$$

Hence,

$$3x = \ln\left(\frac{7}{4}\right)$$

and therefore

$$x = \frac{1}{3} \ln\left(\frac{7}{4}\right)$$

Using a calculator,

$$x \approx 0.187$$

3.5.5 Applications of Exponential Functions

Engineering Application

Populations are often modelled using exponential functions.

$$P = P_0 e^{kt},$$

where P_0 is the initial population and k is the growth rate. If $k > 0$, the population increases exponentially over time.

Engineering Application

Radioactive materials decay according to an exponential law.

$$N = N_0 e^{-kt},$$

where N_0 is the initial quantity and k is a positive constant. The negative exponent causes the quantity to decrease over time.

Engineering Application

Exponential functions frequently arise in engineering applications, including:

- capacitor discharge,
- Newton's law of cooling,
- vibration damping,
- atmospheric pressure models,
- belt tension calculations.

3.5.6 Exercises

Exercise

1. Evaluate the following using a calculator:
 - (a) e^3
 - (b) $5e^{2.5}$
 - (c) $2.8e^{-1.2}$

2. Determine whether the following functions represent exponential growth or exponential decay:
 - (a) $y = 3e^{0.4x}$
 - (b) $y = 7e^{-0.2x}$
 - (c) $y = 2e^{1.5x}$

3. Determine the y -intercept of the following functions:
 - (a) $y = 5e^{0.6x}$
 - (b) $y = 4e^{-x} + 3$
 - (c) $y = 2e^{2x} - 1$

4. Solve the following equations.
 - (a) $e^x = 20$
 - (b) $3e^x = 15$
 - (c) $5e^{2x} - 2 = 18$

3.5 Solutions to Exercise 3.5.6

1. Evaluate the following using a calculator:
 - (a) Using a calculator, $e^3 \approx 20.09$ to 2 decimal places.
 - (b) Using a calculator, $5e^{2.5} = 60.91$ to 2 decimal places.
 - (c) Using a calculator, $2.8e^{-1.2} = 0.84$ to 2 decimal places.

2. Recall:

- If the coefficient of x in the exponent is positive, the function exhibits exponential growth.
- If the coefficient of x in the exponent is negative, the function exhibits exponential decay.

(a) Since $0.4 > 0$, the function represents exponential growth.

(b) Since $-0.2 < 0$, the function represents exponential decay.

(c) Since $1.5 > 0$, the function represents exponential growth.

3. The y -intercept occurs when $x = 0$.

(a) Substituting $x = 0$:

$$\begin{aligned} y &= 5e^{0.6x} \\ &= 5e^{0.6(0)} \\ &= 5e^0 \\ &= 5(1) \\ &= 5 \end{aligned}$$

Therefore, the y -intercept is $(0, 5)$.

(b) Substituting $x = 0$:

$$\begin{aligned} y &= 4e^{-x} + 3 \\ &= 4e^{-0} + 3 \\ &= 4e^0 + 3 \\ &= 4(1) + 3 \\ &= 7 \end{aligned}$$

Therefore, the y -intercept is $(0, 7)$.

(c) Substituting $x = 0$:

$$\begin{aligned}y &= 2e^{2x} - 1 \\&= 2e^{2(0)} - 1 \\&= 2e^0 - 1 \\&= 2(1) - 1 \\&= 1\end{aligned}$$

Therefore, the y -intercept is $(0, 1)$.

4. Solve the following equations:

(a) Taking the natural logarithm of both sides:

$$\begin{aligned}e^x &= 20 \\ \ln(e^x) &= \ln(20) \\ x &= \ln(20) \\ &= 3.00 \text{ to 2 decimal places.}\end{aligned}$$

(b)

$$\begin{aligned}3e^x &= 15 && \text{divide both sides by 3} \\ e^x &= 5 && \text{taking the natural logarithm of both sides} \\ \ln(e^x) &= \ln(5) \\ x &= \ln(5) \\ &= 1.61 \text{ to 2 decimal places}\end{aligned}$$

(c)

$$5e^{2x} - 2 = 18$$

add 2 to both sides

$$5e^{2x} = 20$$

dividing by 5

$$e^{2x} = 4$$

taking the natural logarithm of both sides

$$\ln(e^{2x}) = \ln(4)$$

$$2x = \ln(4)$$

$$x = \frac{\ln(4)}{2}$$

$$= 0.69 \text{ to 2 decimal places}$$